

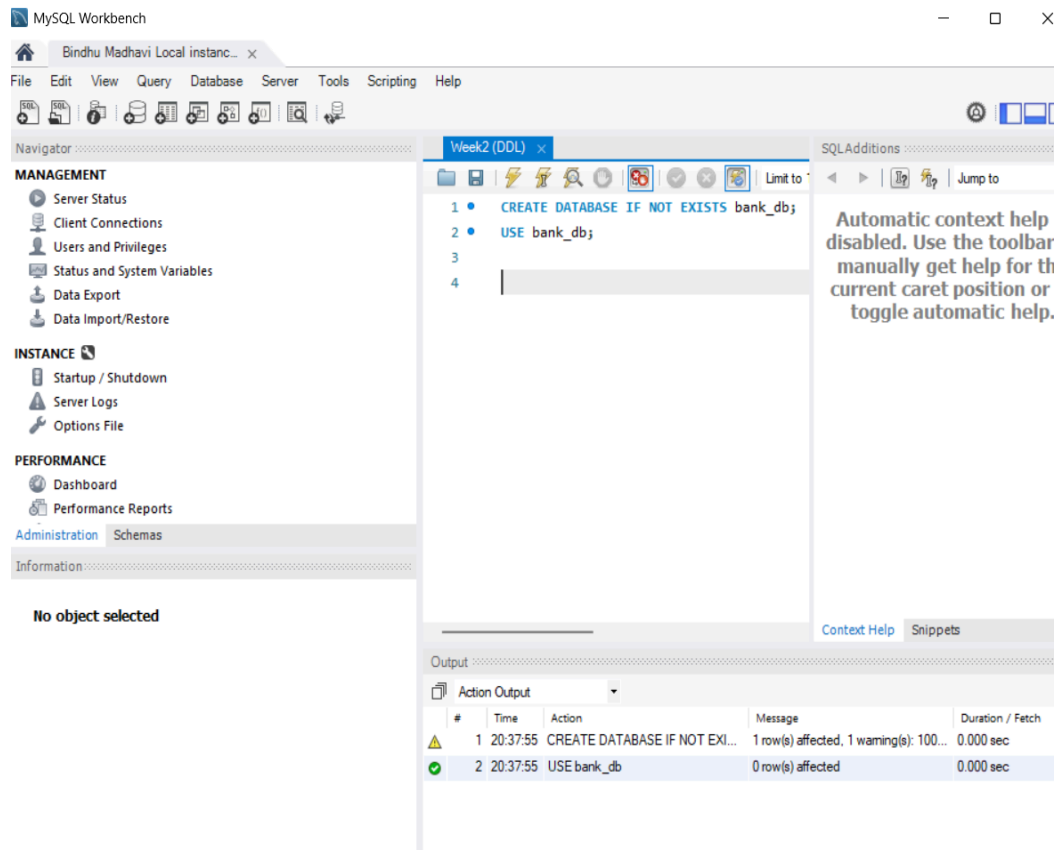
Week-2 (DDL/DCL/TCL)

Name: P. Bindhu Madhavi

Roll Number: 2520030077

DDL Commands

STEP 1: Create Database



STEP 2: Create Table

STEP 3: Alter Table Add Column

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'Navigator' pane with sections for MANAGEMENT, INSTANCE, and PERFORMANCE. The main editor window, titled 'Week2 (DDL)*', contains the following SQL code:

```
1 CREATE DATABASE IF NOT EXISTS bank_db;
2 USE bank_db;
3 CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     branch_name VARCHAR(50),
7     transaction_type VARCHAR(20),
8     amount DECIMAL(10,2),
9     transaction_date DATE
10 );
```

The right sidebar shows the 'SQLAdditions' pane with a message: 'Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.' The bottom pane shows the 'Output' window with the following table:

#	Time	Action	Message	Duration / Fetch
1	20:37:55	CREATE DATABASE IF NOT EXI...	1 row(s) affected, 1 warning(s): 100...	0.000 sec
2	20:37:55	USE bank_db	0 row(s) affected	0.000 sec
3	20:43:04	CREATE TABLE bank_transaction...	0 row(s) affected	0.094 sec

The screenshot shows the MySQL Workbench interface. The main editor window, titled 'Week2 (DDL)*', contains the following SQL code:

```
1 CREATE DATABASE IF NOT EXISTS bank_db;
2 USE bank_db;
3 CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     branch_name VARCHAR(50),
7     transaction_type VARCHAR(20),
8     amount DECIMAL(10,2),
9     transaction_date DATE
10 );
11 ALTER TABLE bank_transactions
12 ADD account_no VARCHAR(20);
13 ALTER TABLE bank_transactions
14 MODIFY customer_name VARCHAR(100);
```

The right sidebar shows the 'SQLAdditions' pane with a message: 'Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.' The bottom pane shows the 'Output' window with the following table:

#	Time	Action	Message	Duration / Fetch
1	20:37:55	CREATE DATABASE IF NOT EXISTS...	1 row(s) affected, 1 warning(s): 1007 C...	0.000 sec
2	20:37:55	USE bank_db	0 row(s) affected	0.000 sec
3	20:43:04	CREATE TABLE bank_transactions (...	0 row(s) affected	0.094 sec
4	20:45:36	ALTER TABLE bank_transactions AD...	0 row(s) affected Records: 0 Duplicat...	0.063 sec
5	20:57:05	ALTER TABLE bank_transactions MO...	0 row(s) affected Records: 0 Duplicat...	0.141 sec

STEP 4: Rename Table

MySQL Workbench

Bindhu Madhavi Local instanc...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges**
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Week2 (DDL)*

```

2 • USE bank_db;
3 • CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     branch_name VARCHAR(50),
7     transaction_type VARCHAR(20),
8     amount DECIMAL(10,2),
9     transaction_date DATE
10 );
11 • ALTER TABLE bank_transactions
12     ADD account_no VARCHAR(20);
13 • ALTER TABLE bank_transactions
14     MODIFY customer_name VARCHAR(100);
15 • RENAME TABLE bank_transactions TO custom
16 • RENAME TABLE customer_transactions TO be
17

```

SQLAdditions

Automatic context help disabled. Use the toolbar manually get help for the current caret position or toggle automatic help

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	20:37:55	CREATE DATABASE IF NOT EXISTS...	1 row(s) affected, 1 warning(s): 1007 C...	0.000 sec
2	20:37:55	USE bank_db	0 row(s) affected	0.000 sec
3	20:43:04	CREATE TABLE bank_transactions (...	0 row(s) affected	0.094 sec
4	20:45:36	ALTER TABLE bank_transactions AD...	0 row(s) affected Records: 0 Duplicat...	0.063 sec
5	20:57:05	ALTER TABLE bank_transactions MO...	0 row(s) affected Records: 0 Duplicat...	0.141 sec
6	21:01:04	RENAME TABLE bank_transactions ...	0 row(s) affected	0.047 sec
7	21:01:04	RENAME TABLE customer_transactio...	0 row(s) affected	0.156 sec

STEP 5: Create Backup Table

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

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- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Query 1 Week2 Week2 (DDL)*

```

2 • USE bank_db;
3 • CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     branch_name VARCHAR(50),
7     transaction_type VARCHAR(20),
8     amount DECIMAL(10,2),
9     transaction_date DATE
10 );
11 • ALTER TABLE bank_transactions
12 ADD account_no VARCHAR(20);
13 • ALTER TABLE bank_transactions
14 MODIFY customer_name VARCHAR(100);
15 • RENAME TABLE bank_transactions TO custom
16 • RENAME TABLE customer_transactions TO be
17 • CREATE TABLE bank_backup AS
18 SELECT * FROM bank_transactions;

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	20:37:55	CREATE DATABASE IF NOT EXISTS...	1 row(s) affected, 1 warning(s): 1007 C...	0.000 sec
2	20:37:55	USE bank_db	0 row(s) affected	0.000 sec
3	20:43:04	CREATE TABLE bank_transactions (...	0 row(s) affected	0.094 sec
4	20:45:36	ALTER TABLE bank_transactions AD...	0 row(s) affected Records: 0 Duplicat...	0.063 sec
5	20:57:05	ALTER TABLE bank_transactions MO...	0 row(s) affected Records: 0 Duplicat...	0.141 sec
6	21:01:04	RENAME TABLE bank_transactions ...	0 row(s) affected	0.047 sec
7	21:01:04	RENAME TABLE customer_transactio...	0 row(s) affected	0.156 sec
8	21:03:46	CREATE TABLE bank_backup AS S...	0 row(s) affected Records: 0 Duplicat...	0.063 sec

Object Info Session

STEP 6: Insert Sample Data

STEP 7: Update Account Numbers

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator

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INSTANCE

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- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Query 1 Week2 Week2 (DDL)*

```

16 • RENAME TABLE customer_transactions TO be
17 • CREATE TABLE bank_backup AS
18 SELECT * FROM bank_transactions;
19 • INSERT INTO bank_transactions
20 (txn_id,customer_name,branch_name,transa
21 VALUES
22 (101,'Ravi','Hyderabad','Deposit',5000,'
23 (102,'Sita','Hyderabad','Withdrawal',200
24 (103,'Kiran','Vijayawada','Deposit',1200
25 (104,'Anil','Vizag','Deposit',8000,'2024
26 (105,'Priya','Hyderabad','Withdrawal',35
27 (106,'Ramesh','Vizag','Deposit',15000,'2
28 (107,'Keerthi','Vijayawada','Withdrawal'
29 (108,'Rahul','Hyderabad','Deposit',9000,
30 (109,'Sneha','Vizag','Withdrawal',4000,'
31 (110,'Madhu','Vijayawada','Deposit',1100

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	20:37:55	USE bank_db	0 row(s) affected	0.000 sec
3	20:43:04	CREATE TABLE bank_transactions ...	0 row(s) affected	0.094 sec
4	20:45:36	ALTER TABLE bank_transactions A...	0 row(s) affected Records: 0 Duplic...	0.063 sec
5	20:57:05	ALTER TABLE bank_transactions ...	0 row(s) affected Records: 0 Duplic...	0.141 sec
6	21:01:04	RENAME TABLE bank_transactions...	0 row(s) affected	0.047 sec
7	21:01:04	RENAME TABLE customer_transact...	0 row(s) affected	0.156 sec
8	21:03:46	CREATE TABLE bank_backup AS ...	0 row(s) affected Records: 0 Duplic...	0.063 sec
9	21:06:38	INSERT INTO bank_transactions (tx...	10 row(s) affected Records: 10 Duplic...	0.016 sec

Object Info Session

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator

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INSTANCE

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- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Query 1 Week2 Week2 (DDL)* SQLAdditions

```

15 • RENAME TABLE bank_transactions TO custom
16 • RENAME TABLE customer_transactions TO be
17 • CREATE TABLE bank_backup AS
18 • SELECT * FROM bank_transactions;
19 • INSERT INTO bank_transactions
20 (txn_id,customer_name,branch_name,transa
21 VALUES
22 (101,'P.Bindhu Madhavi','Hyderabad','Dep
23 (102,'P.Bindhu Madhavi','Hyderabad','Wit
24 (103,'P.Bindhu Madhavi','Vijayawada','De
25 (104,'P..Bindhu Madhavi','Vizag','Deposi
26 (105,'Priya','Hyderabad','Withdrawal',35
27 (106,'Ramesh','Vizag','Deposit',15000,'2
28 (107,'Keerthi','Vijayawada','Withdrawal'
29 (108,'Rahul','Hyderabad','Deposit',9000,
30 (109,'Sneha','Vizag','Withdrawal',4000,
31 (110,'Madhu','Vijayawada','Deposit',1100

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
13	21:25:17	UPDATE bank_transactions SET...	1 row(s) affected Rows matched: ...	0.000 sec
14	21:25:17	UPDATE bank_transactions SET...	1 row(s) affected Rows matched: ...	0.000 sec
15	21:25:17	UPDATE bank_transactions SET...	1 row(s) affected Rows matched: ...	0.000 sec
16	21:25:17	UPDATE bank_transactions SET...	1 row(s) affected Rows matched: ...	0.000 sec
17	21:25:17	UPDATE bank_transactions SET...	1 row(s) affected Rows matched: ...	0.000 sec
18	21:25:17	UPDATE bank_transactions SET...	1 row(s) affected Rows matched: ...	0.016 sec
19	21:25:17	UPDATE bank_transactions SET...	1 row(s) affected Rows matched: ...	0.000 sec
20	21:26:21	INSERT INTO bank_transaction...	1 row(s) affected	0.000 sec
21	21:26:24	INSERT INTO bank_transaction...	Error Code: 1063 Database error	0.000 sec

STEP 8: Insert

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

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INSTANCE

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- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Query 1 Week2 Week2 (DDL)* SQLAdditions

```

29 (108,'Rahul','Hyderabad','Deposit',9000,
30 (109,'Sneha','Vizag','Withdrawal',4000,
31 (110,'Madhu','Vijayawada','Deposit',1100
32 • UPDATE bank_transactions SET account_no
33 • UPDATE bank_transactions SET account_no
34 • UPDATE bank_transactions SET account_no
35 • UPDATE bank_transactions SET account_no
36 • UPDATE bank_transactions SET account_no
37 • UPDATE bank_transactions SET account_no
38 • UPDATE bank_transactions SET account_no
39 • UPDATE bank_transactions SET account_no
40 • UPDATE bank_transactions SET account_no
41 • UPDATE bank_transactions SET account_no
42 • INSERT INTO bank_transactions
43 VALUES
44 (111,'Venu','Vizag','Deposit',7000,'2024

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message
13	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...
14	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...
15	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...
16	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...
17	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...
18	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...
19	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...
20	21:12:20	INSERT INTO bank_transactions V...	1 row(s) affected

Object Info Session

STEP 9: Update

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

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- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Query 1 Week2 Week2 (DDL)*

```

32 • UPDATE bank_transactions SET account_no=
33 • UPDATE bank_transactions SET account_no=
34 • UPDATE bank_transactions SET account_no=
35 • UPDATE bank_transactions SET account_no=
36 • UPDATE bank_transactions SET account_no=
37 • UPDATE bank_transactions SET account_no=
38 • UPDATE bank_transactions SET account_no=
39 • UPDATE bank_transactions SET account_no=
40 • UPDATE bank_transactions SET account_no=
41 • UPDATE bank_transactions SET account_no=
42 • INSERT INTO bank_transactions
43 VALUES
44 (111,'Venu','Vizag','Deposit',7000,'2024-09-01')
45 • UPDATE bank_transactions
46 SET amount=5000
47 WHERE txn_id=105;

```

SQL Additions

Automatic context help disabled. Use the tool manually get help for current caret position toggle automatic help

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 14	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...	0.000 sec
✓ 15	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...	0.000 sec
✓ 16	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...	0.000 sec
✓ 17	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...	0.000 sec
✓ 18	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...	0.015 sec
✓ 19	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...	0.000 sec
✓ 20	21:12:20	INSERT INTO bank_transactions V...	1 row(s) affected	0.016 sec
✓ 21	21:14:26	UPDATE bank_transactions SET a...	1 row(s) affected Rows matched: 1 ...	0.000 sec

Object Info Session

STEP 10: Delete

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Query 1 Week2 Week2 (DDL)*

```

34 • UPDATE bank_transactions SET account_no=
35 • UPDATE bank_transactions SET account_no=
36 • UPDATE bank_transactions SET account_no=
37 • UPDATE bank_transactions SET account_no=
38 • UPDATE bank_transactions SET account_no=
39 • UPDATE bank_transactions SET account_no=
40 • UPDATE bank_transactions SET account_no=
41 • UPDATE bank_transactions SET account_no=
42 • INSERT INTO bank_transactions
43 VALUES
44 (111,'Venu','Vizag','Deposit',7000,'2024-09-01')
45 • UPDATE bank_transactions
46 SET amount=5000
47 WHERE txn_id=105;
48 • DELETE FROM bank_transactions
49 WHERE txn_id=111;

```

SQL Additions

Automatic context help disabled. Use the tool manually get help for current caret position toggle automatic help

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 15	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...	0.000 s
✓ 16	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...	0.000 s
✓ 17	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...	0.000 s
✓ 18	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...	0.015 s
✓ 19	21:09:29	UPDATE bank_transactions SET ac...	1 row(s) affected Rows matched: 1 ...	0.000 s
✓ 20	21:12:20	INSERT INTO bank_transactions V...	1 row(s) affected	0.016 s
✓ 21	21:14:26	UPDATE bank_transactions SET a...	1 row(s) affected Rows matched: 1 ...	0.000 s
✓ 22	21:15:59	DELETE FROM bank_transactions ...	1 row(s) affected	0.015 s

Object Info Session

STEP 11: Select

MySQL Workbench

Bindhu Madhavi Local instanc...

File Edit View Query Database Server Tools Scripting Help

Navigator

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INSTANCE

- Startup / Shutdown
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- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Query 1 Week2 Week2 (DDL)*

```

50 WHERE txn_id=111;
51 SELECT * FROM bank_transactions;
52 SELECT * FROM bank_transactions
53 WHERE transaction_type='Deposit';
54 SELECT * FROM bank_transactions
55 ORDER BY amount DESC;
56

```

Automatic context help disabled. Use the toolbar manually get help for current caret position, toggle automatic help

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount
106	Ramesh	Vizag	Deposit	1000
103	P.Bindhu Madhavi	Vijayawada	Deposit	500
110	Madhu	Vijayawada	Deposit	200
108	Rahul	Hyderabad	Deposit	150
104	P.Bindhu Madhavi	Vizag	Deposit	100
101	P.Bindhu Madhavi	Hyderabad	Deposit	50
105	Priya	Hyderabad	Withdrawal	100
109	Sneha	Vizag	Withdrawal	50

bank_transactions 2 bank Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
18	21:25:17	UPDATE bank_transactions SET...	1 row(s) affected Rows matched...	0.016 sec
19	21:25:17	UPDATE bank_transactions SET...	1 row(s) affected Rows matched...	0.000 sec
20	21:26:21	INSERT INTO bank_transactions...	1 row(s) affected	0.000 sec
21	21:26:24	INSERT INTO bank_transactions...	Error Code: 1062 Duplicate entry...	0.000 sec
22	21:32:23	UPDATE bank_transactions SET...	1 row(s) affected Rows matched...	0.000 sec
23	21:32:53	DELETE FROM bank_transactions...	1 row(s) affected	0.000 sec
24	21:33:40	SELECT * FROM bank_transactions...	10 row(s) returned	0.000 sec / 0.000 sec
25	21:33:40	SELECT * FROM bank_transactions...	6 row(s) returned	0.000 sec / 0.000 sec
26	21:33:40	SELECT * FROM bank_transactions...	10 row(s) returned	0.000 sec / 0.000 sec

DCL COMMANDS

STEP 1: Create User - Auditor1 & Branch Manager

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator

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INSTANCE

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- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Query 1 Week2 Week2 (DDL)* DCL Commands*

```

1 CREATE USER IF NOT EXISTS 'Auditor1'@'localhost'
2 IDENTIFIED BY 'audit123';
3
4 CREATE USER IF NOT EXISTS 'BranchManager'@'localhost'
5 IDENTIFIED BY 'manager123';
6

```

Output

Action Output

#	Time	Action	Message
20	21:26:21	INSERT INTO bank_transactions VALUES (111,'Venu','Vizag','Deposit',7000,'2024-01-18','A...'	1 row(s) affected
21	21:26:24	INSERT INTO bank_transactions VALUES (111,'Venu','Vizag','Deposit',7000,'2024-01-18','A...'	Error Code: 1062. Duplicate entry '111' for key 'bank_transa
22	21:32:23	UPDATE bank_transactions SET amount=5000 WHERE bn_id=105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings:
23	21:32:53	DELETE FROM bank_transactions WHERE bn_id=111	1 row(s) affected
24	21:33:40	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned
25	21:33:40	SELECT * FROM bank_transactions WHERE transaction_type=Deposit LIMIT 0, 1000	6 row(s) returned
26	21:33:40	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned
27	21:40:04	CREATE USER IF NOT EXISTS 'Auditor1'@'localhost' IDENTIFIED BY 'audit123'	0 row(s) affected
28	21:40:04	CREATE USER IF NOT EXISTS 'BranchManager'@'localhost' IDENTIFIED BY 'manager123'	0 row(s) affected

Object Info Session

STEP 2: Grant SELECT Permission

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

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INSTANCE

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- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Query 1 Week2 Week2 (DDL)* DCL Commands*

```

1 CREATE USER IF NOT EXISTS 'Auditor1'@'localhost'
2 IDENTIFIED BY 'audit123';
3
4 CREATE USER IF NOT EXISTS 'BranchManager'@'localhost'
5 IDENTIFIED BY 'manager123';
6
7 GRANT SELECT
8 ON bookflow_db.bank_transactions
9 TO 'Auditor1'@'localhost';
10
11 GRANT ALL PRIVILEGES
12 ON bookflow_db.bank_transactions
13 TO 'BranchManager'@'localhost';

```

Automatic console disabled. Use the manual get current caret position to toggle auto

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration
22	21:32:23	UPDATE bank_transactions SET amount=5000 WHERE bn_id=105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings:	0.000 s
23	21:32:53	DELETE FROM bank_transactions WHERE bn_id=111	1 row(s) affected	0.000 s
24	21:33:40	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned	0.000 s
25	21:33:40	SELECT * FROM bank_transactions WHERE transaction_type=Deposit LIMIT 0, 1000	6 row(s) returned	0.000 s
26	21:33:40	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned	0.000 s
27	21:40:04	CREATE USER IF NOT EXISTS 'Auditor1'@'localhost' IDENTIFIED BY 'audit123'	0 row(s) affected	0.093 s
28	21:40:04	CREATE USER IF NOT EXISTS 'BranchManager'@'localhost' IDENTIFIED BY 'manager123'	0 row(s) affected	0.016 s
29	21:41:45	GRANT SELECT ON bookflow_db.bank_transactions TO 'Auditor1'@'localhost';	0 row(s) affected	0.015 s
30	21:41:45	GRANT ALL PRIVILEGES ON bookflow_db.bank_transactions TO 'BranchManager'@'localhost';	0 row(s) affected	0.000 s

Object Info Session

STEP 3: Check Current User

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator

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INSTANCE

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- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

Week2 Week2 (DDL)* DCL Commands*

```

1 CREATE USER IF NOT EXISTS 'Auditor1'@'localhost'
2 IDENTIFIED BY 'audit123';
3
4 CREATE USER IF NOT EXISTS 'BranchManager'@'localhost'
5 IDENTIFIED BY 'manager123';
6 GRANT SELECT
7 ON bookflow_db.bank_transactions
8 TO 'BranchManager'@'localhost';

```

Result Grid

Filter Rows:

Result Grid

Form Editor

Result 1 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration
23	21:32:53	DELETE FROM bank_transactions	1 row(s) affected	0.000 sec
24	21:33:40	SELECT * FROM bank_transactions	10 row(s) returned	0.000 sec
25	21:33:40	SELECT * FROM bank_transactions	6 row(s) returned	0.000 sec
26	21:33:40	SELECT * FROM bank_transactions	10 row(s) returned	0.000 sec
27	21:40:04	CREATE USER IF NOT EXISTS	0 row(s) affected	0.093 sec
28	21:40:04	CREATE USER IF NOT EXISTS	0 row(s) affected	0.016 sec
29	21:41:45	GRANT SELECT ON bookflow_db.bank_transactions	0 row(s) affected	0.015 sec
30	21:41:45	GRANT ALL PRIVILEGES ON bookflow_db.bank_transactions	0 row(s) affected	0.000 sec
31	21:49:14	SELECT CURRENT_USER()	1 row(s) returned	0.000 sec

Object Info Session

Automatic content disabled. Use the manual caret position to toggle automatic content.

STEP 4: Revoke SELECT Permission & Revoke ALL PRIVILEGES

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

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Administration Schemas

Information

No object selected

Week2 (DDL)* DCL Commands* Weeks2(TCL)

```

10 ON bookflow_db.bank_transactions
11 TO 'BranchManager'@'localhost';
12 SELECT CURRENT_USER();
13 REVOKE SELECT
14 ON bookflow_db.bank_transactions
15 FROM 'Auditor1'@'localhost';
16

```

Result Grid

Filter Rows:

Result Grid

Form Editor

Result 1 x

Read Only Context Help Snippets

Output

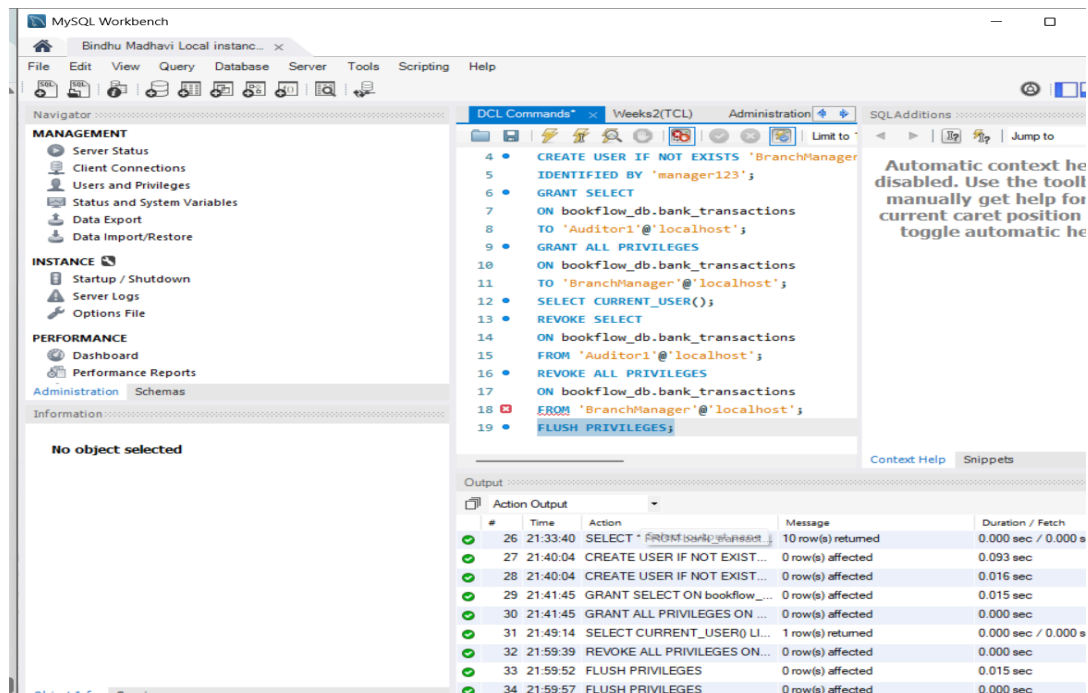
Action Output

#	Time	Action	Message	Duration
23	21:32:53	DELETE FROM bank_transactions	1 row(s) affected	0.000 se
24	21:33:40	SELECT * FROM bank_transactions	10 row(s) returned	0.000 se
25	21:33:40	SELECT * FROM bank_transactions	6 row(s) returned	0.000 se
26	21:33:40	SELECT * FROM bank_transactions	10 row(s) returned	0.000 se
27	21:40:04	CREATE USER IF NOT EXISTS	0 row(s) affected	0.093 se
28	21:40:04	CREATE USER IF NOT EXISTS	0 row(s) affected	0.016 se
29	21:41:45	GRANT SELECT ON bookflow_db.bank_transactions	0 row(s) affected	0.015 se
30	21:41:45	GRANT ALL PRIVILEGES ON bookflow_db.bank_transactions	0 row(s) affected	0.000 se
31	21:49:14	SELECT CURRENT_USER()	1 row(s) returned	0.000 se

Object Info Session

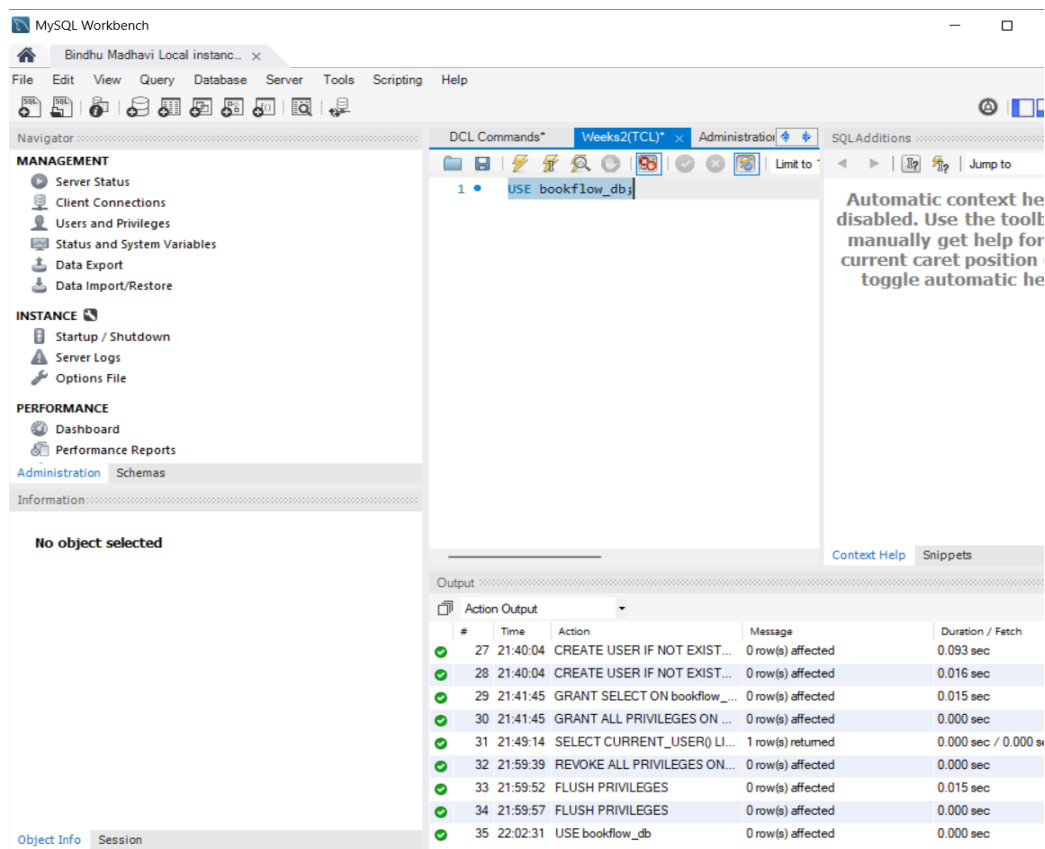
Automatic content disabled. Use the manual caret position to toggle automatic content.

STEP 5: Refresh Privileges



TCL COMMANDS

STEP 1: Select the Database



STEP 2: Disable Auto Commit && Update Record

MySQL Workbench interface showing a transaction in progress. The SQL editor contains the following commands:

```

1 • USE bookflow_db;
2 • SET AUTOCOMMIT = 0;
3 • UPDATE bank_transactions
4   SET amount = 6000
5   WHERE txn_id = 101;
6
7 • COMMIT;

```

The output table shows the execution of these commands:

#	Time	Action	Message	Duration / Fetch
30	21:41:45	GRANT ALL PRIVILEGES ON ...	0 row(s) affected	0.000 sec
31	21:49:14	SELECT CURRENT_USER() LI...	1 row(s) returned	0.000 sec / 0.000 sec
32	21:59:39	REVOKE ALL PRIVILEGES ON...	0 row(s) affected	0.000 sec
33	21:59:52	FLUSH PRIVILEGES	0 row(s) affected	0.015 sec
34	21:59:57	FLUSH PRIVILEGES	0 row(s) affected	0.000 sec
35	22:02:31	USE bookflow_db	0 row(s) affected	0.000 sec
36	22:04:17	SET AUTOCOMMIT = 0	0 row(s) affected	0.000 sec
37	22:04:35	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec
38	22:04:35	COMMIT	0 row(s) affected	0.000 sec

STEP 3: Commit the Transaction & Create a Save Point

MySQL Workbench interface showing the transaction completed and a save point created. The SQL editor contains the following commands:

```

1 • USE bookflow_db;
2 • SET AUTOCOMMIT = 0;
3 • UPDATE bank_transactions
4   SET amount = 6000
5   WHERE txn_id = 101;
6
7 • COMMIT;
8 • SAVEPOINT Before_Update;
9

```

The output table shows the execution of these commands:

#	Time	Action	Message	Duration / Fetch
31	21:49:14	SELECT CURRENT_USER() LI...	1 row(s) returned	0.000 sec /
32	21:59:39	REVOKE ALL PRIVILEGES ON...	0 row(s) affected	0.000 sec
33	21:59:52	FLUSH PRIVILEGES	0 row(s) affected	0.015 sec
34	21:59:57	FLUSH PRIVILEGES	0 row(s) affected	0.000 sec
35	22:02:31	USE bookflow_db	0 row(s) affected	0.000 sec
36	22:04:17	SET AUTOCOMMIT = 0	0 row(s) affected	0.000 sec
37	22:04:35	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec
38	22:04:35	COMMIT	0 row(s) affected	0.000 sec
39	22:06:18	SAVEPOINT Before_Update	0 row(s) affected	0.000 sec

STEP 4: Update Another Record

MySQL Workbench

Bindhu Madhavi Local instanc...

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Information

No object selected

Weeks2(TCL)*

```

1 • USE bookflow_db;
2 • SET AUTOCOMMIT = 0;
3 • UPDATE bank_transactions
4   SET amount = 6000
5   WHERE txn_id = 101;
6
7 • COMMIT;
8 • SAVEPOINT Before_Update;
9 • UPDATE bank_transactions
10  SET amount = 99999
11  WHERE txn_id = 102;

```

Automatic context disabled. Use the to manually get help current caret position toggle automatic

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
32	21:59:39	REVOKE ALL PRIVILEGES ON...	0 row(s) affected	0.000 sec
33	21:59:52	FLUSH PRIVILEGES	0 row(s) affected	0.015 sec
34	21:59:57	FLUSH PRIVILEGES	0 row(s) affected	0.000 sec
35	22:02:31	USE bookflow_db	0 row(s) affected	0.000 sec
36	22:04:17	SET AUTOCOMMIT = 0	0 row(s) affected	0.000 sec
37	22:04:35	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec
38	22:04:35	COMMIT	0 row(s) affected	0.000 sec
39	22:06:18	SAVEPOINT Before_Update	0 row(s) affected	0.000 sec
40	22:07:41	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec

Object Info Session

STEP 5: Rollback To Save Point

MySQL Workbench

Bindhu Madhavi Local instanc...

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Administration Schemas

Information

No object selected

Weeks2(TCL)*

```

1 • USE bookflow_db;
2 • SET AUTOCOMMIT = 0;
3 • UPDATE bank_transactions
4   SET amount = 6000
5   WHERE txn_id = 101;
6
7 • COMMIT;
8 • SAVEPOINT Before_Update;
9 • UPDATE bank_transactions
10  SET amount = 99999
11  WHERE txn_id = 102;
12 • ROLLBACK TO Before_Update;

```

Automatic context disabled. Use the too manually get help for current caret position toggle automatic h

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
33	21:59:52	FLUSH PRIVILEGES	0 row(s) affected	0.015 sec
34	21:59:57	FLUSH PRIVILEGES	0 row(s) affected	0.000 sec
35	22:02:31	USE bookflow_db	0 row(s) affected	0.000 sec
36	22:04:17	SET AUTOCOMMIT = 0	0 row(s) affected	0.000 sec
37	22:04:35	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec
38	22:04:35	COMMIT	0 row(s) affected	0.000 sec
39	22:06:18	SAVEPOINT Before_Update	0 row(s) affected	0.000 sec
40	22:07:41	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec
41	22:09:15	ROLLBACK TO Before_Update	0 row(s) affected	0.000 sec

Object Info Session

STEP 6: Full TCL Example

MySQL Workbench interface showing a TCL script execution. The script is titled "Weeks2(TCL)" and contains the following SQL statements:

```
11 WHERE txn_id = 102;
12 ROLLBACK TO Before_Update;
13 UPDATE bank_transactions
14 SET amount = 7000
15 WHERE txn_id = 101;
16
17 SAVEPOINT SP1;
18
19 UPDATE bank_transactions
20 SET amount = 9000
21 WHERE txn_id = 102;
22
23 ROLLBACK TO SP1;
24
25 COMMIT;
26
```

The output table shows the execution results:

#	Time	Action	Message	Duration / Fetch
38	22:04:35	COMMIT	0 row(s) affected	0.000 sec
39	22:06:18	SAVEPOINT Before_Update	0 row(s) affected	0.000 sec
40	22:07:41	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec
41	22:09:15	ROLLBACK TO Before_Update	0 row(s) affected	0.000 sec
42	22:11:04	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.016 sec
43	22:11:04	SAVEPOINT SP1	0 row(s) affected	0.000 sec
44	22:11:04	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec
45	22:11:04	ROLLBACK TO SP1	0 row(s) affected	0.000 sec
46	22:11:04	COMMIT	0 row(s) affected	0.000 sec

STEP 7: Output

MySQL Workbench

Bindhu Madhavi Local instance...

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Administration Schemas

Information

No object selected

Weeks2(TCL)*

```
21 WHERE txn_id = 102;
22
23 ROLLBACK TO SP1;
24
25 COMMIT;
26 SELECT * FROM bank_transactions
27 WHERE txn_id IN (101,102);
```

Result Grid

txn_id	customer_name	branch_name	trans
101	P.Bindhu Madhavi	Hyderabad	Depos
102	P.Bindhu Madhavi	Hyderabad	Withd

transactions 1

Output

Action Output

#	Time	Action	Message	Duration / Fetch
39	22:06:18	SAVEPOINT Before_Update	0 row(s) affected	0.000 sec
40	22:07:41	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec
41	22:09:15	ROLLBACK TO Before_Update	0 row(s) affected	0.000 sec
42	22:11:04	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.016 sec
43	22:11:04	SAVEPOINT SP1	0 row(s) affected	0.000 sec
44	22:11:04	UPDATE bank_transactions SE...	1 row(s) affected Rows matched...	0.000 sec
45	22:11:04	ROLLBACK TO SP1	0 row(s) affected	0.000 sec
46	22:11:04	COMMIT	0 row(s) affected	0.000 sec
47	22:12:17	SELECT * FROM bank_transact	2 row(s) returned	0.000 sec / 0.00

Automatic context disabled. Use the tool manually get help from current caret position toggle automatic